RAJALAKSHMI ENGINEERING COLLEGE

Ram Karthik Subramanian 251001206-2025-IT

R2

Started on	Thursday, 27 August 2026, 11:00 PM
State	Finished
Completed on	Thursday, 27 August 2026, 11:02 PM
Time taken	2 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int arr[100005];

// Divide and conquer: find index of first 0 (binary search for transition point)
int findFirstZero(int low, int high) {
    if (low > high) {
        return -1; // no zero found in this range
    }

    if (low == high) {
        return (arr[low] == 0) ? low : -1;
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == 1) {
        // First zero (if any) lies in the right half
        return findFirstZero(mid + 1, high);
    } else {
        // arr[mid] == 0, so first zero is at mid or somewhere in left half
        int leftResult = findFirstZero(low, mid - 1);
        if (leftResult != -1) {
            return leftResult;
        } else {
            return mid;
        }
    }
}

int main() {
    int m;
    scanf("%d", &m);

    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    if (m == 0) {
        printf("0\n");
        return 0;
    }

    int firstZeroIndex = findFirstZero(0, m - 1);

    int countZeroes;
    if (firstZeroIndex == -1) {
        countZeroes = 0; // no zero found, all 1s
    } else {
        countZeroes = m - firstZeroIndex;
    }

```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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